A

PYTHON INSTALLATION

This appendix covers how to install Python, as well as the external modules and code used in the book. Installation for the hardware projects using Raspberry Pi is covered in **Appendix B**. The projects in this book have been tested with Python 3.9.

Installing Source Code for the Book's Projects

You can download source code for the book's projects from ***https://github.com/mkvenkit/pp2e***. Use the Download ZIP option at this site to retrieve the code.

Once you download and extract the code, you need to add the path to the *common* folder in the downloaded code (generally *pp-master/common/*) to your `PYTHONPATH` environment variable so that modules can find and use these Python files.

On Windows, you can do this by creating a `PYTHONPATH` environment variable or adding to one if it already exists. On macOS, you can add this line to your *.profile* file in your home directory (or create a file there if needed):

export PYTHONPATH=$PYTHONPATH:`path_to_common_folder`

Fill in the path to the common folder as appropriate.

Linux users can do something similar to macOS, in their *.bashrc*, *.bash_profile*, or *.cshrc/.login* files as appropriate. Use the `echo $SHELL` command to see the default shell.

Installing Python and Python Modules

I recommend using the Anaconda distribution of Python to run the book's projects, since it already comes with most of the Python modules you'll need. This section covers the installation process on Windows, macOS, and Linux.

Windows

Visit ***https://www.anaconda.com*** and download Anaconda Distribution for Windows. Once the installation is complete, bring up the Anaconda prompt (type **Anaconda prompt** in the search bar), which you'll use to run your programs. Just `cd` into the book's code directory and you're ready to go.

It's also useful to add the location of Anaconda and its supporting files into your `Path` environment variable (type **Edit Environment Variables** in the search bar):

`C:\Users\mahes\anaconda3`

`C:\Users\mahes\anaconda3`\Scripts

`C:\Users\mahes\anaconda3`\Library\bin

In this case, my Anaconda installation directory was `C:\Users\mahes\anaconda3\`. Modify this as necessary.

Installing GLFW

For the OpenGL-based 3D graphics projects in this book, you need the GLFW library, which you can download at ***http://www.glfw.org***. On Windows, after you install GLFW, set a `GLFW_LIBRARY` environment variable (type **Edit Environment Variables** in the search bar) to the full path of the installed *glfw3.dll* so that your Python binding for GLFW can find this library. The path will look something like *C:\glfw-3.0.4.bin.WIN32\lib-msvc120\glfw3.dll*.

To use GLFW with Python, you use a module called `pyglfw`, which consists of a single Python file called *glfw.py*. You don't need to install `pyglfw` because it comes with the source code for the book; it's in the *common* directory. Just in case you need to install a more recent version, here is the source: ***https://github.com/rougier/pyglfw***.

You also need to ensure that your graphics card drivers are installed on your computer. This is a good thing in general since many programs (especially games) make use of the graphics processing unit (GPU).

Installing Additional Modules

You need to install a few additional modules that aren't part of the standard Anaconda Distribution. Run the following commands at the Anaconda prompt:

```
conda install -c anaconda pyaudio

conda install -c anaconda pyopengl
```

macOS

Visit ***https://www.anaconda.com*** and download Anaconda Distribution for macOS. Once it's installed, bring up a Terminal application window and enter `which python`. The output should point to the version of Anaconda Python. If not, add the path manually to your *.profile* file. For example:

```
export
PYTHONPATH=your_anaconda_install_dir_path:$PYTHONPATH
```

Fill in the path to your Anaconda installation directory as appropriate.

Installing GLFW

For the OpenGL-based 3D graphics projects in this book, you need the GLFW library, which you can download at *http://www.glfw.org*. Choose the macOS **Pre-compiled Binaries** option and copy the downloaded folder to your *Home* folder. In my case, for example, my *Home* folder is */Users/mahesh/*.

Now you need to add the following to your *.profile*, making changes to your path as appropriate:

```
export GLFW_LIBRARY=/Users/mahesh/glfw-3.3.8.bin.MACOS/lib-universal/libglfw.3.dylib
```

You may get a security warning when you first try to run a program that uses GLFW. You need to allow it to run in System Preferences under Security.

Installing Additional Modules

Next, you need to install some additional modules that aren't part of the standard Anaconda Distribution. In a Terminal window, run the following commands:

```
conda install -c anaconda pyaudio
```

```
conda install -c anaconda pyopengl
```

Linux

Linux usually comes with Python, as well as all the development tools needed to build the required packages, built in. So you don't need to install Anaconda Python. On most Linux distributions, you should be able to use `pip3` to get the packages required for the book. You can install a package using `pip3` like this:

```
sudo pip3 install matplotlib
```

The other way to install a package is to download the module source distribution for it, which is usually in a *.gz* or *.zip* file. Once you unzip these files into a folder, you can then install them as follows:

```
sudo python setup.py install
```

Use one of these methods for each package needed for the book.